

Name: Srinjoy Chakraborty

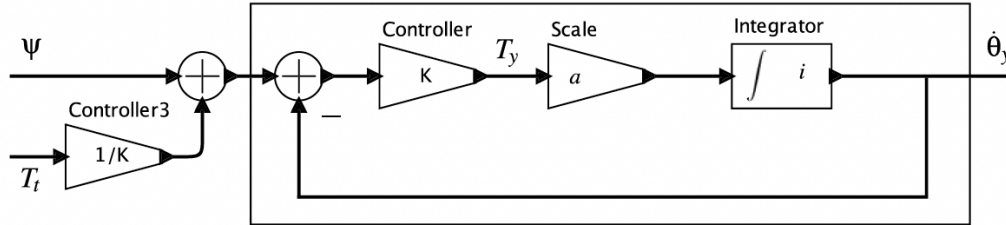
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Homework Assignment 1

Deadline: September 11, 2026

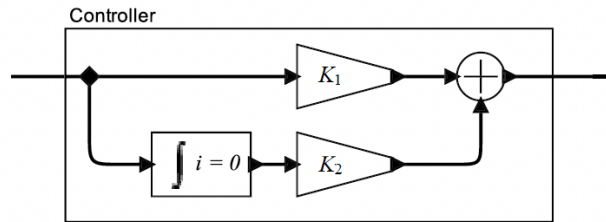
Total: 60 marks**Problem 1.** (10 points)

(a) Construct a Simulink model of the helicopter control system shown in the figure below.



Choose some reasonable parameters and plot the actual angular velocity as a function of time, assuming that the desired angular velocity is zero, $\psi(t) = 0$, and that the top-rotor torque is non-zero, $T_t(t) = bu(t)$. Give your plot for three different values of K and discuss how the behavior varies with K .

(b) Modify the model of part (a) to replace the Controller (the simple scale-by- K actor) with the alternative controller shown in the figure below.



This alternative controller is called a proportional-integrator (PI) controller. It has two parameters K_1 and K_2 . Experiment with the values of these parameters, give some plots of the behavior with the same inputs as in part (a), and discuss the behavior of this controller in contrast to the one of part (a).

Problem 1: Helicopter Control System**(a) Proportional Controller Behavior**

A Simulink model was constructed to test the proportional scale-by- K controller using three parallel systems to evaluate different values of the gain K . The desired angular velocity was set to $\psi(t) = 0$, and the top-rotor torque $T_t(t)$ was modeled as a unit Step input. To observe the response, the following parameters were chosen:

- **Case 1:** $K = 4$, Scale $a = 2$, Torque Input Gain = $1/4$
- **Case 2:** $K = 5$, Scale $a = 2$, Torque Input Gain = $1/5$
- **Case 3:** $K = 6$, Scale $a = 2$, Torque Input Gain = $1/6$

Discussion: Because the forward path contains a single integrator from the plant dynamics, the closed-loop system is a first-order system. The response to the torque disturbance is a smooth exponential curve with no overshoot. The final steady-state angular velocity settles at $1/K$. Therefore, as the proportional gain K increases, the steady-state error decreases (approaching the desired zero state more closely), and the system settles faster due to a smaller time constant.

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Homework Assignment 1

Deadline: September 11, 2026

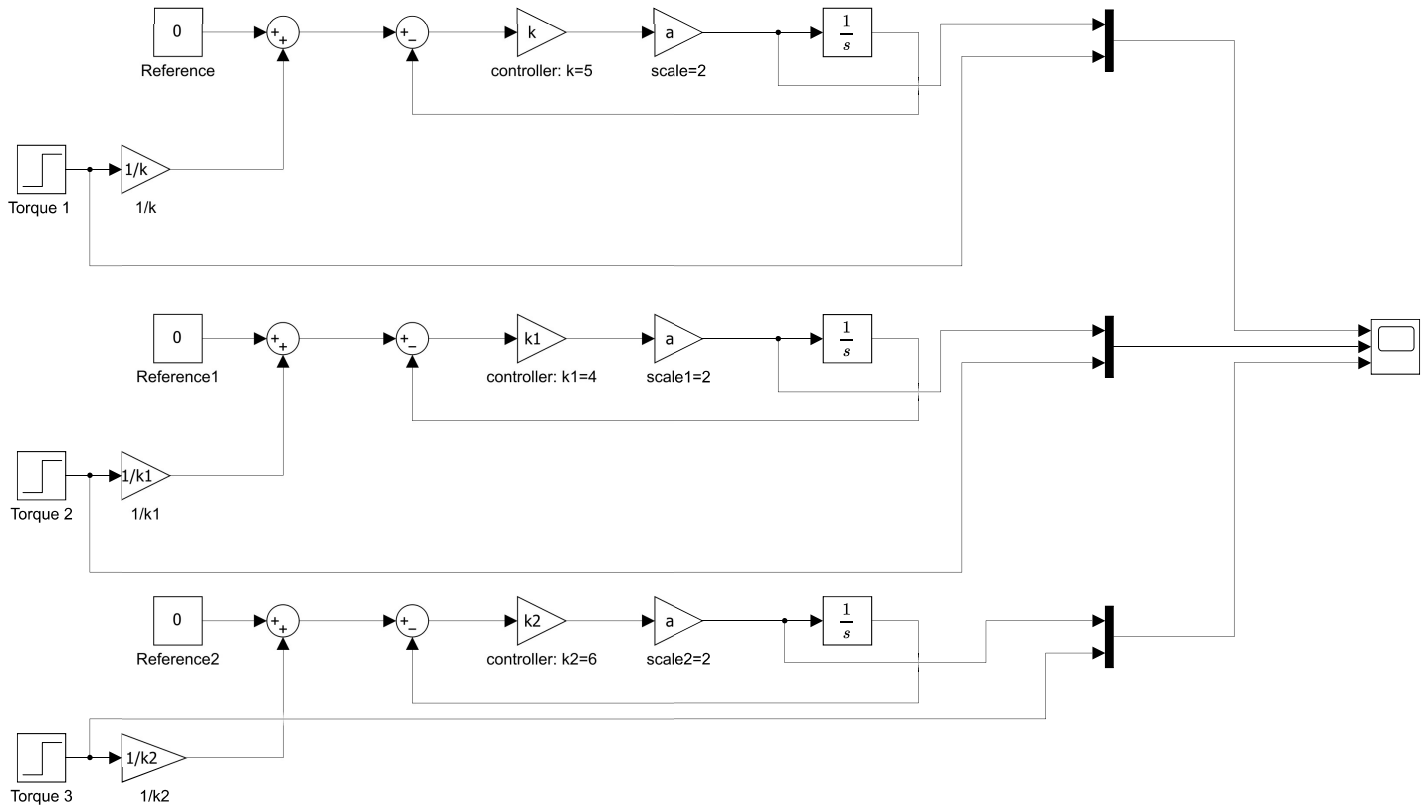


Figure 1: Simulink layout for Part (a) testing three different K values in parallel.

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CS637 Embedded and Cyber-Physical SystemsRoll No: 251040638
e.g. 170001Dept.: EE
e.g. CSE

Homework Assignment 1

Deadline: September 11, 2026

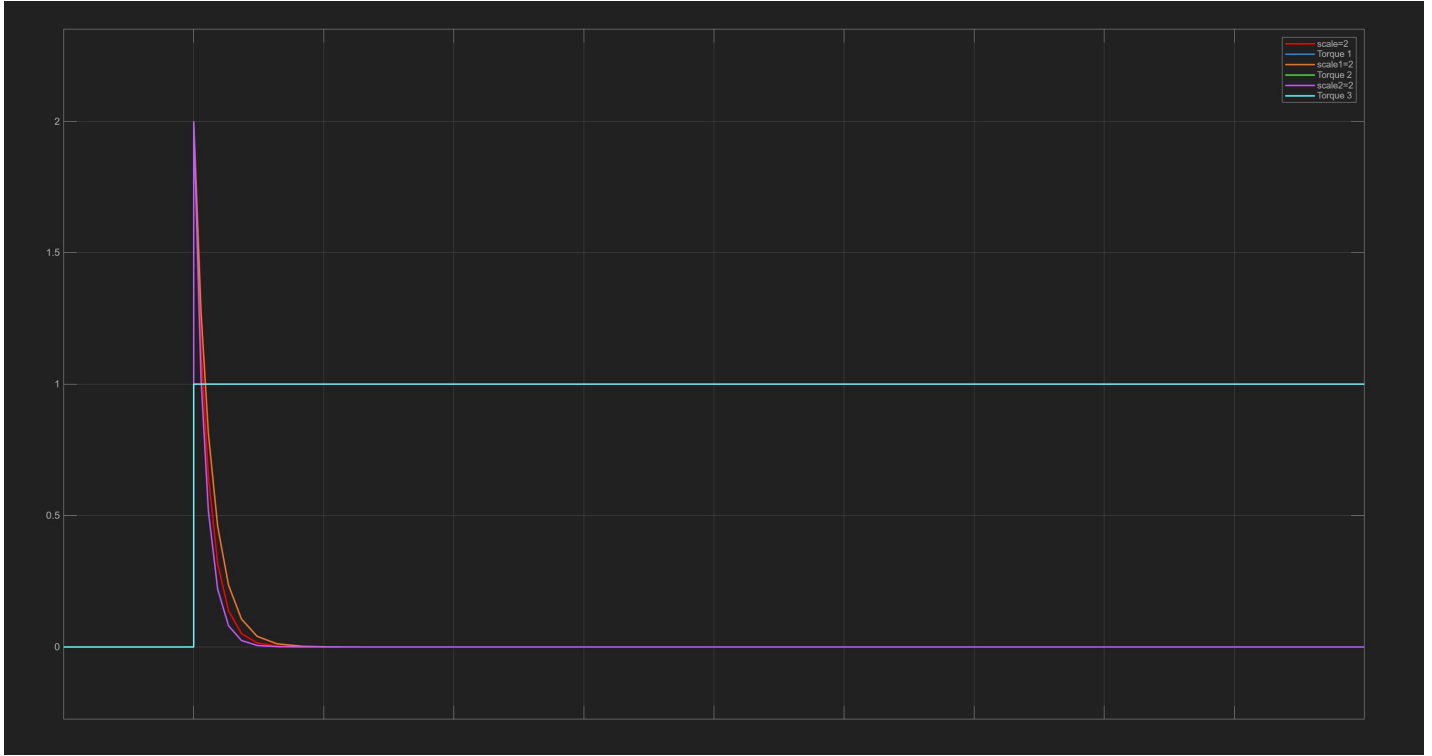


Figure 2: Angular velocity $\dot{\theta}_y$ over time for Part (a), showing first-order exponential settling at exactly $1/K$.

(b) Proportional-Integrator (PI) Controller Behavior

The proportional controller was replaced with a PI controller featuring a proportional gain K_1 and an integral gain K_2 . The torque input gain was updated to $1/K_1$ to properly scale the disturbance reference[cite: 1]. The following three test cases were evaluated in parallel[cite: 1]:

- **Case 1 (Top Loop):** $K_1 = 4$, $K_2 = 2$, Scale $a = 2$, Torque Gain = $1/4$ [cite: 1]
- **Case 2 (Middle Loop):** $K_1 = 2$, $K_2 = 4$, Scale $a = 2$, Torque Gain = $1/2$ [cite: 1]
- **Case 3 (Bottom Loop):** $K_1 = 5$, $K_2 = 10$, Scale $a = 2$, Torque Gain = $1/5$ [cite: 1]

Contrast with Part (a): Introducing the integral action ($1/s$) into the controller transforms the closed-loop dynamics into a second-order system. Unlike the first-order exponential curve in Part (a), the PI controller exhibits an underdamped response characterized by an initial overshoot (and undershoot below zero) before settling. Because the effective scaled reference is $1/K_1$, the integrator successfully drives the steady-state value to exactly $1/K_1$. Tuning the parameters reveals that higher integral gains (K_2) generally cause more aggressive oscillations, while adjusting K_1 dictates the final steady-state amplitude.

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Homework Assignment 1

Deadline: September 11, 2026

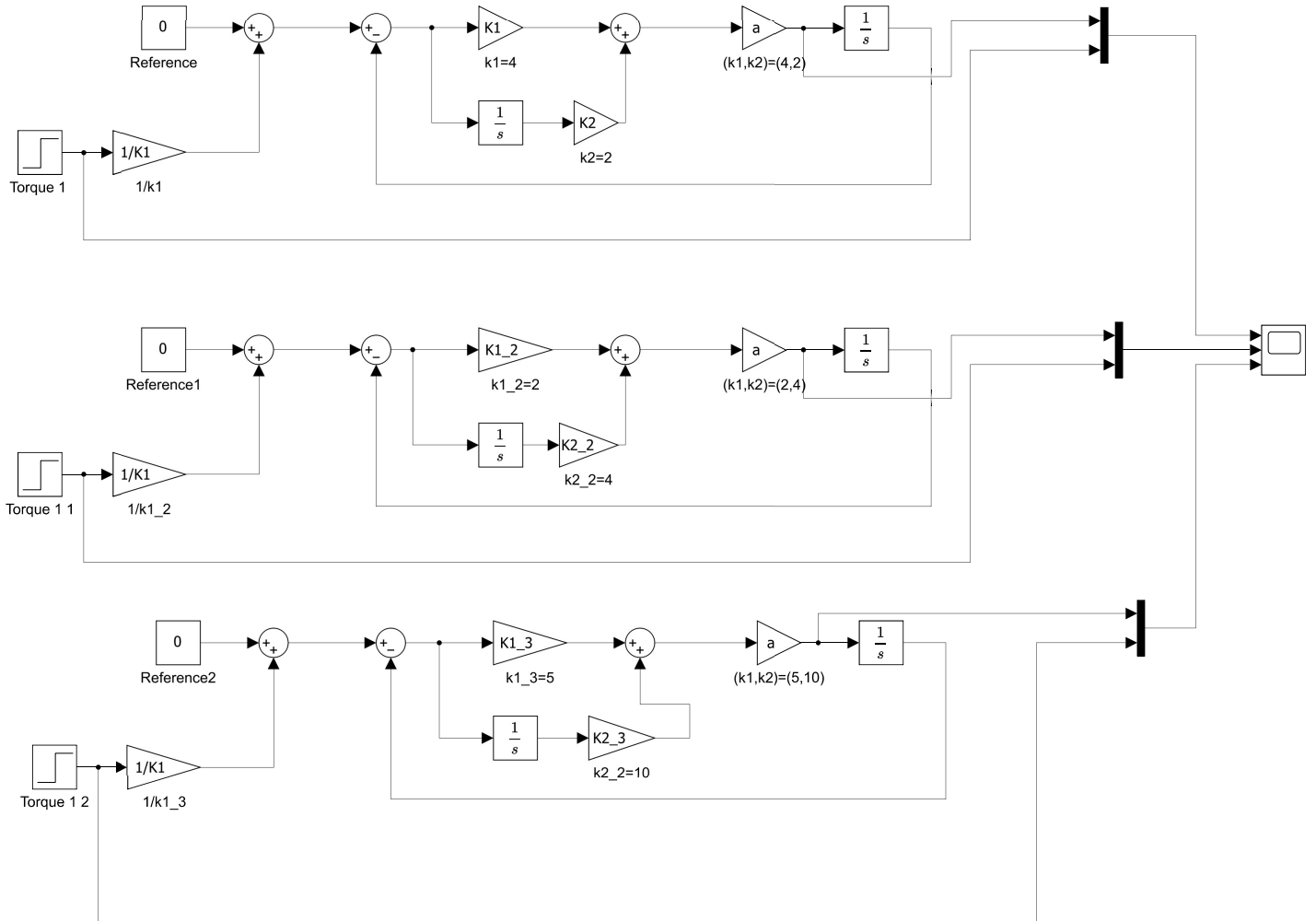


Figure 3: Modified Simulink layout replacing the simple gain with the PI controller architecture, configured for negative feedback[cite: 1].

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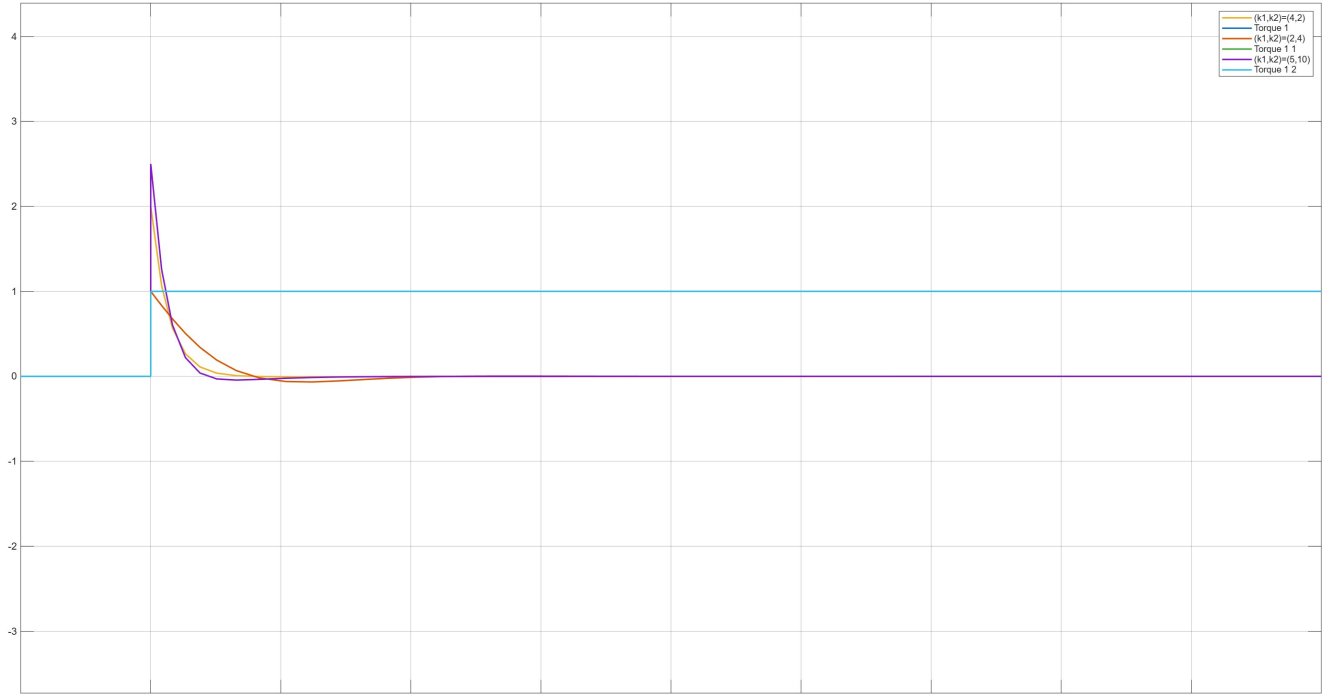
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Deadline: September 11, 2026

Figure 4: Angular velocity $\dot{\theta}_y$ over time for Part (b). The curves exhibit second-order dynamics (overshoot/wobble) before settling exactly at their respective $1/K_1$ steady-state values.

Name: Srinjoy Chakraborty

Indian Institute of Technology Kanpur
CS637 Embedded and Cyber-Physical SystemsRoll No: 251040638
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e.g. CSE

Homework Assignment 1

Deadline: September 11, 2026

Problem 2. (10 points)

The states of the linearized model of a vehicle steering system represent the lateral deviation of the vehicle from the x-axis and the angle between the vehicle axis and the x -axis. The output of the linearized model is only the first state. Construct a Simulink model for the vehicle steering system with its controller that includes an observer. The dynamics are available in Example 6.4 and Example 7.3 in [AM09]. Apply a sinusoidal signal as the reference trajectory that specifies the desired deviation of the vehicle from the x-axis with time. Plot the output (lateral deviation of the vehicle from the x-axis) with time.

[AM09] K. J. Astrom and R. M. Murray. Feedback Systems: An Introduction for Scientists and Engineers. Princeton University Press, 2009.

http://www.cds.caltech.edu/~murray/books/AM08/pdf/am08-complete_22Feb09.pdf.

Problem 2: Vehicle Steering System with Observer

Based on the normalized kinematic bicycle model [AM09], the continuous-time plant matrices describing lateral deviation are:

$$A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} \gamma \\ 1 \end{bmatrix}, \quad C = [1 \quad 0], \quad D = 0$$

1. Controller Variables:

To design a controller that tracks a reference trajectory r and achieves a desired characteristic polynomial $p(s) = s^2 + 2\zeta_c\omega_c s + \omega_c^2$, the feedback control law $u = -Kx + k_r r$ is applied. Comparing the closed-loop characteristic polynomial with the desired $p(s)$ yields the state feedback gains:

$$k_1 = \omega_c^2, \quad k_2 = 2\zeta_c\omega_c - \gamma\omega_c^2$$

To ensure a DC gain of 1 for steady-state tracking, the feedforward gain is set to $k_r = k_1 = \omega_c^2$. Assuming $\omega_c = 1$ rad/s, $\zeta_c = 0.707$, and $\gamma = 0.5$, the variables evaluate to $K = [1 \quad 0.914]$ and $k_r = 1$.

2. Observer Variables:

An observer is introduced to estimate the internal states, designed to be twice as fast as the controller ($\omega_o = 2\omega_c = 2$ rad/s, $\zeta_o = 0.707$). Matching the observer characteristic polynomial $\det(sI - A + LC) = s^2 + l_1 s + l_2$ yields the observer gains:

$$l_1 = 2\zeta_o\omega_o = 2.828, \quad l_2 = \omega_o^2 = 4 \implies L = \begin{bmatrix} 2.828 \\ 4 \end{bmatrix}$$

For implementation in Simulink via a standard State-Space block[cite: 3], the internal matrices are defined as: $A_{obs} = A - LC$, $B_{obs} = [B \quad L]$, and $C_{obs} = I_{2 \times 2}$ to output the estimated state vector $\hat{x}$ into the feedback gain K .

Name: Srinjoy Chakraborty

Roll No: 251040638
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Homework Assignment 1

Deadline: September 11, 2026

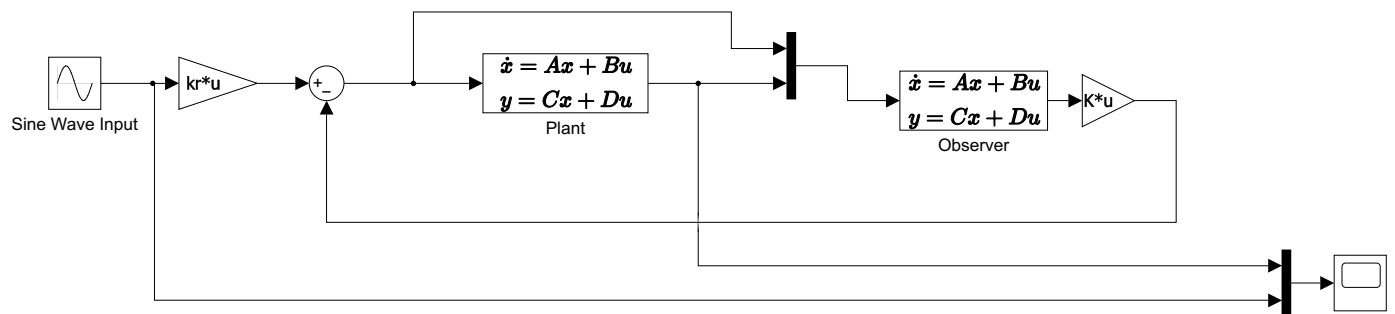


Figure 1: Simulink layout of the vehicle steering system. The analytical feedback laws and observer dynamics are mapped to the Gain and State-Space blocks[cite: 3].

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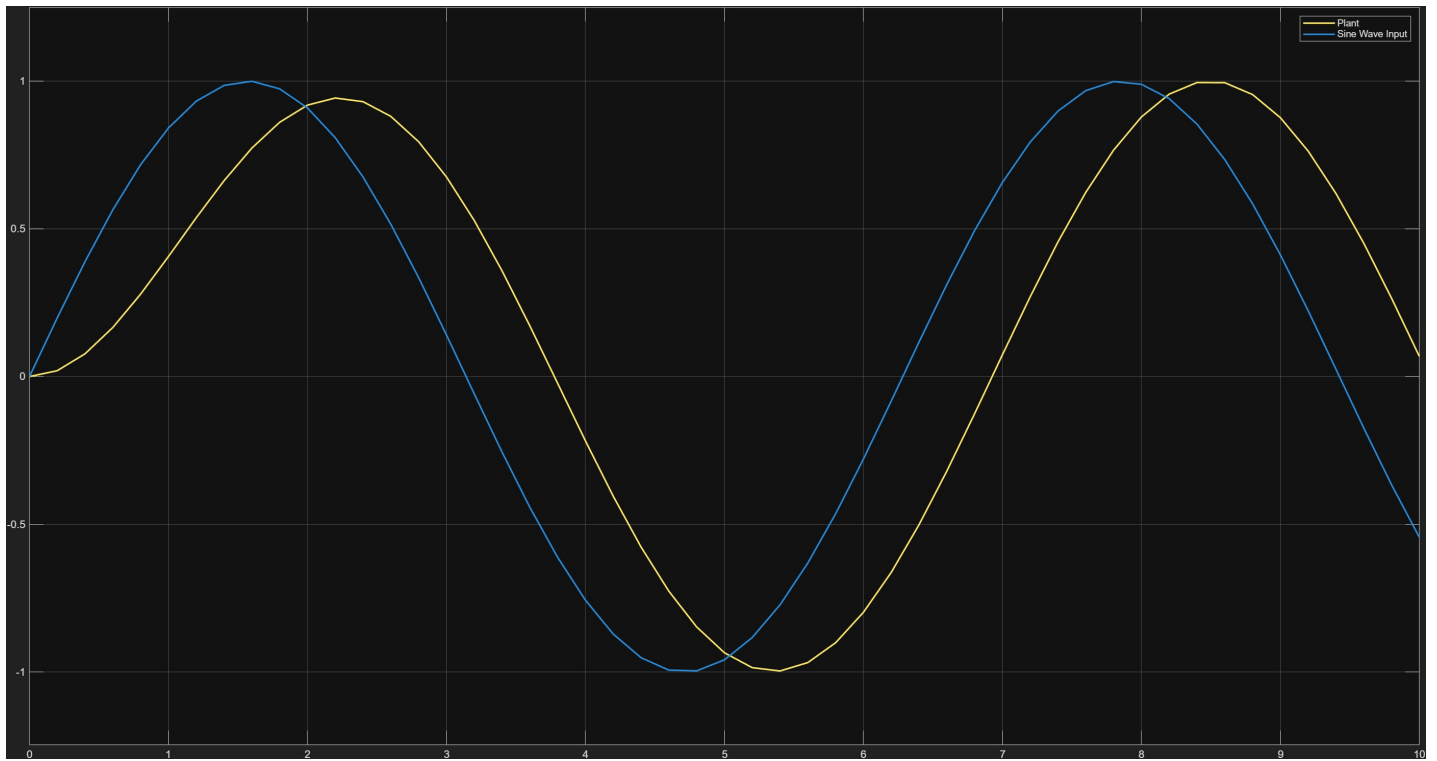
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Deadline: September 11, 2026

Figure 2: System response plotting the actual lateral deviation y tracking the sinusoidal reference trajectory $r(t) = \sin(0.5t)$ using the analytically derived observer.

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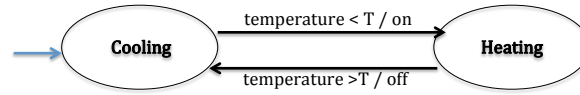
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e.g. CSE

Homework Assignment 1

Deadline: September 11, 2026

Problem 3. (10 points)

Consider the following model for a thermostat system.



The thermostat has been designed to maintain the temperature of a room at $T^{\circ}\text{C}$. The model has two states: *cooling* and *heating*. When the system is in the *cooling* state and the temperature of the room goes below $T^{\circ}\text{C}$, the system generates a signal to switch on a heater and moves to the *heating* state. When the temperature of the room goes over $T^{\circ}\text{C}$, the system generates a signal to switch off the heater and moves to the *cooling* state.

- Represent the system as an actor that takes the current temperature as input and produces a signal to control the heater. The actor uses the set point T as a parameter.
- Identify a design problem in the model.
- Provide two different remedies to address the problem.
- Compare your proposed two solutions in terms of ease of implementation and guarantees on the system behavior.

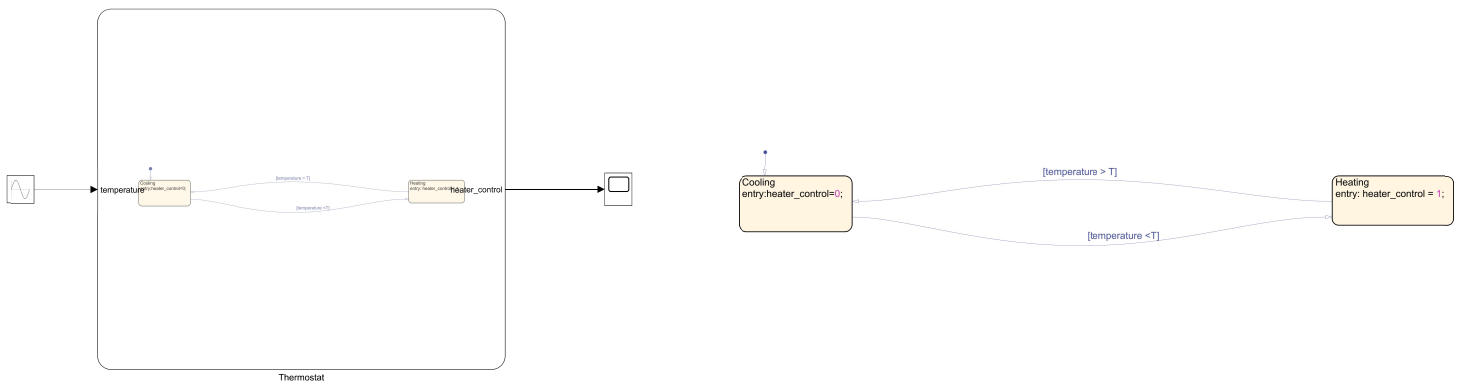
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CS637 Embedded and Cyber-Physical SystemsRoll No: 251040638
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e.g. CSEHomework Assignment 1
Deadline: September 11, 2026**(a) Actor Representation**

The thermostat system is represented as a Stateflow chart actor in Simulink. The actor takes `temperature` as a continuous input signal and produces `heater_control` as a discrete output signal.

Initializations: The set point parameter T is defined in the MATLAB workspace (e.g., using `T = Simulink.Parameter; T.Value = 24;`). The variable T is assigned the Scope of “Parameter” in the Stateflow Symbols pane, allowing the chart to read it during execution.

Connections: On the Simulink canvas, a Sine Wave generator is connected to the `temperature` input port to simulate room temperature fluctuations. The `heater_control` output port is wired to a Scope block alongside the temperature input to visualize the state transitions over time.

**(b) Design Problem**

The primary design flaw in this model is **chattering** (or Zeno behavior). Because there is only a single absolute threshold (T), any minor sensor noise or rapid micro-fluctuations around $T^\circ\text{C}$ will cause the system to transition back and forth continuously. This high-frequency toggling can cause severe physical damage to the heater's mechanical relays and drastically reduce its lifespan.

Name: Srinjoy Chakraborty

Roll No: 251040638
e.g. 170001Dept.: EE
e.g. CSEIndian Institute of Technology Kanpur
CS637 Embedded and Cyber-Physical Systems

Homework Assignment 1

Deadline: September 11, 2026

(c) Proposed Remedies

To resolve the chattering issue, the following two remedies can be implemented:

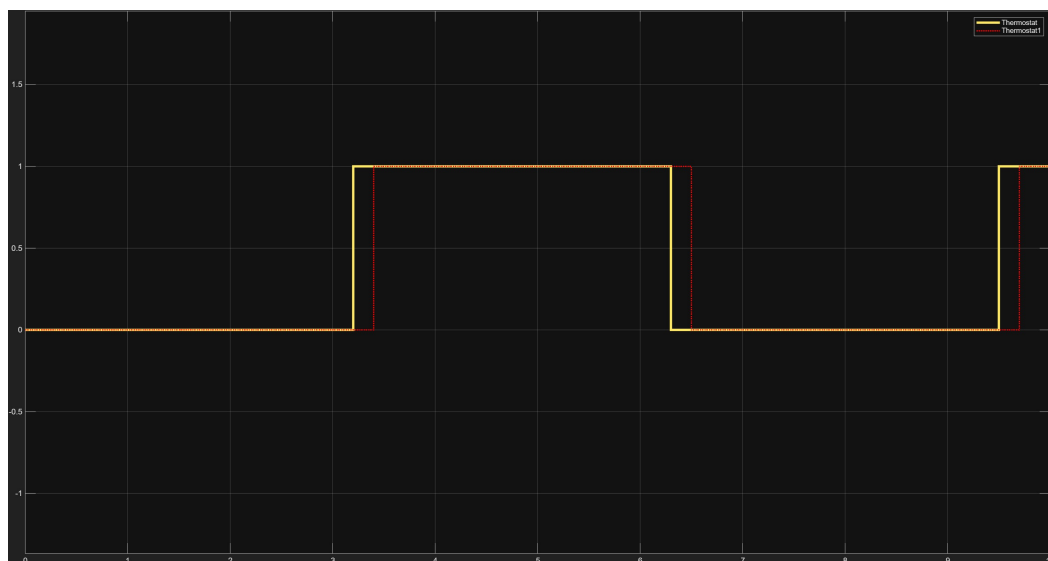
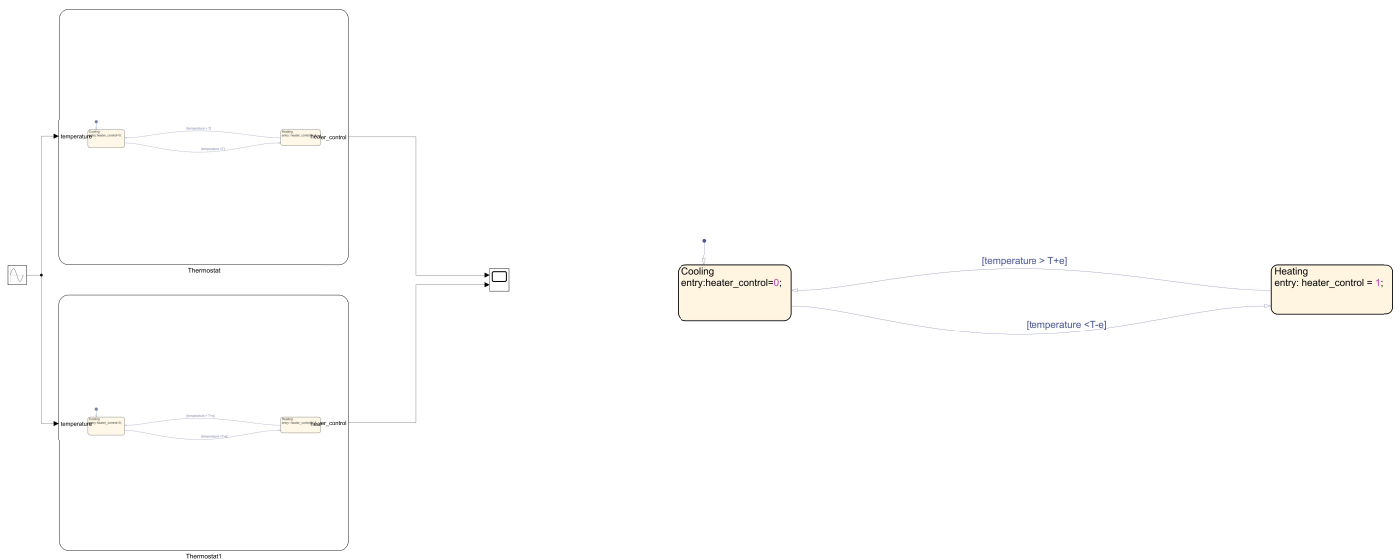
1. **Hysteresis (Spatial Thresholding):** Introduce an error margin or tolerance, e . The system is modified to turn the heater on at $T - e$ and off at $T + e$. This creates a "deadband" that safely absorbs small temperature noise.
2. **Temporal Delay (Minimum Dwell Time):** Enforce a strict duration timer using temporal logic (e.g., `after(5, sec)`). Once the heater switches state, it is forced to remain in that state for a fixed minimum duration before it is allowed to transition again, regardless of temperature fluctuations.

Name: Srinjoy Chakraborty

Indian Institute of Technology Kanpur
CS637 Embedded and Cyber-Physical SystemsRoll No: 251040638
e.g. 170001Dept.: EE
e.g. CSEHomework Assignment 1
Deadline: September 11, 2026**(d) Comparison of Solutions**

Ease of Implementation: Hysteresis is very simple to implement, as it only requires altering the algebraic logic on the transition arrows (e.g., changing $[\text{temperature} < T]$ to $[\text{temperature} < T - e]$). The temporal delay is slightly more complex, requiring absolute-time temporal operators and potentially hierarchical sub-states to manage the timers cleanly.

Guarantees on System Behavior: Hysteresis guarantees stability against noise bounded within the $2e^\circ\text{C}$ band, but if the room temperature itself swings violently, rapid switching could still theoretically occur. The temporal delay, however, provides a *strict, absolute guarantee* on the physical hardware's maximum switching frequency, ensuring the heater is never toggled faster than the defined dwell time under any circumstances.



Name: Srinjoy Chakraborty

Indian Institute of Technology Kanpur
CS637 Embedded and Cyber-Physical Systems
Homework Assignment 1Roll No: 251040638
e.g. 170001Dept.: EE
e.g. CSE

Deadline: September 11, 2026

Problem 4. (10 points)

Construct a timed automation model of a thermostat where the heater will be switched on when the room temperature is less than 22C and switched off when the temperature is above 26C. We also have the following additional requirement. Before switching on or off, it will generate a “beep” sound every second for 5 seconds as a warning. That means once the system senses that the temperature is less than 22C, it will generate the warning first, and then the heater will be switched on. Similarly, a warning sound will be issued before switching off the heater after the temperature reaches 26C.

The timed automaton is implemented using a Simulink Stateflow chart with four primary states to manage the temperature logic and the required 5-second warnings:

- **Cooling:** Heater and beep are OFF. When `temperature < 22`, the system transitions to `Warn_Heating`.
- **Warn_Heating:** Heater remains OFF. A sub-state toggles the `beep` signal every 1 second. After exactly 5 seconds (using the `after(5, sec)` operator), it transitions to `Heating`.
- **Heating:** Heater is ON and beep is OFF. When `temperature > 26`, the system transitions to `Warn_Cooling`.
- **Warn_Cooling:** Heater remains ON. The `beep` toggles every 1 second for 5 seconds before transitioning back to `Cooling`, which finally turns the heater OFF.

A Sine Wave block simulates continuous room temperature changes, while the `heater` and `beep` outputs are routed to a Scope to verify the temporal logic and exact timing delays.

Name: Srinjoy Chakraborty

Roll No: 251040638
e.g. 170001Dept.: EE
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CS637 Embedded and Cyber-Physical Systems

Homework Assignment 1

Deadline: September 11, 2026

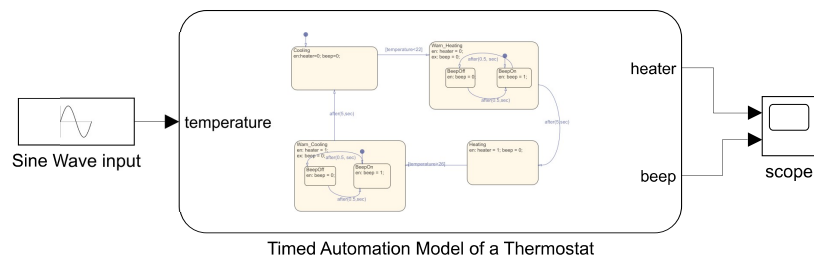


Figure 1: Main Simulink layout showing the input, Stateflow block, and Scope.

Name: Srinjoy Chakraborty

Roll No: 251040638
e.g. 170001Dept.: EE
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Homework Assignment 1

Deadline: September 11, 2026

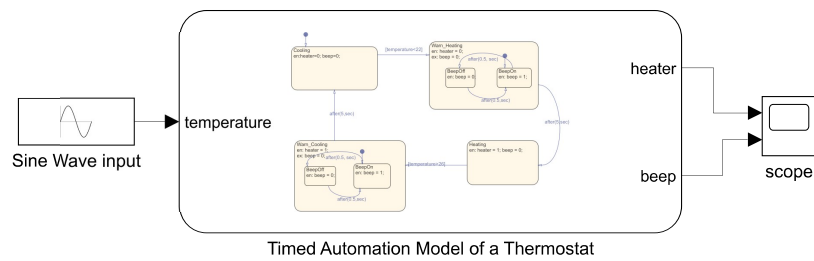


Figure 2: Stateflow chart detailing the timed automaton and warning sub-states.

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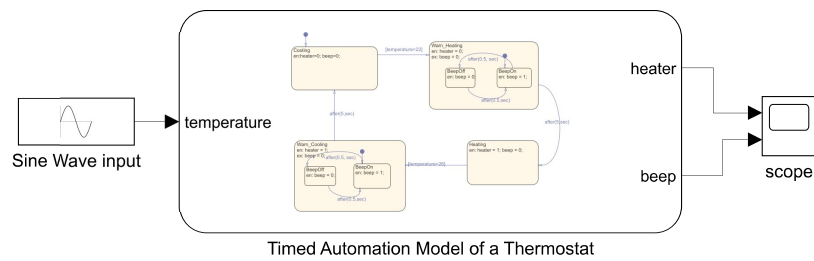
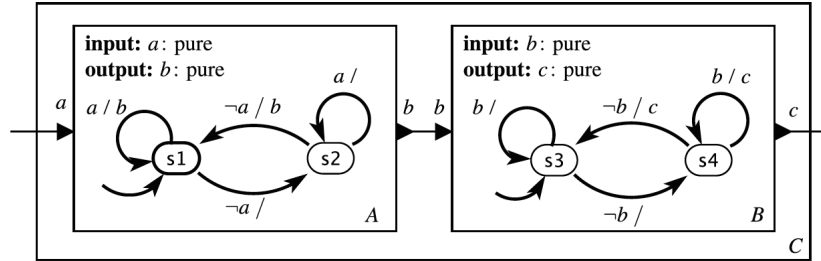
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CS637 Embedded and Cyber-Physical SystemsRoll No: 251040638
e.g. 170001Dept.: EE
e.g. CSEHomework Assignment 1
Deadline: September 11, 2026

Figure 3: Scope plot verifying the temperature thresholds and 5-second warning delays.

Name: Srinjoy Chakraborty

Indian Institute of Technology Kanpur
CS637 Embedded and Cyber-Physical SystemsRoll No: 251040638
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e.g. CSEHomework Assignment 1
Deadline: September 11, 2026**Problem 5.** (10 points)

Consider the following synchronous composition of two state machines A and B.



Construct a single state machine C representing the composition. Which states of the composition are unreachable?

Problem 5: Synchronous Composition of State Machines

To construct a single state machine C representing the synchronous composition of machines A and B , we define the combined state space as $S_C = \{(s1, s3), (s1, s4), (s2, s3), (s2, s4)\}$. The initial state is $(s1, s3)$. In this synchronous composition, Machine A reacts to the pure input a , and its output b (present or absent) instantaneously becomes the input to Machine B, which evaluates it to produce the final output c .

Composition Transition Table:

Current State	Input a	Intermediate b	Next State	Output c
s1_s3	1	present	s1_s3	0
s1_s3	0	present	s2_s3	0
s1_s4	1	present	s1_s4	1
s1_s4	0	present	s2_s4	1
s2_s3	1	absent ($\neg b$)	s2_s4	1
s2_s3	0	absent ($\neg b$)	s1_s4	1
s2_s4	1	absent ($\neg b$)	s2_s3	0
s2_s4	0	absent ($\neg b$)	s1_s3	0

Reachability Analysis:

Zero states are unreachable. By tracing the transitions from the initial state s1_s3, we can prove that a valid sequence of inputs exists to reach every state in the composed machine. Applying an input of $a = 0$ transitions the system to s2_s3. From s2_s3, an input of $a = 1$ reaches s2_s4, while an input of $a = 0$ reaches s1_s4. Thus, the entire state space is accessible.

Name: Srinjoy Chakraborty

Roll No: 251040638
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CS637 Embedded and Cyber-Physical Systems

Homework Assignment 1

Deadline: September 11, 2026

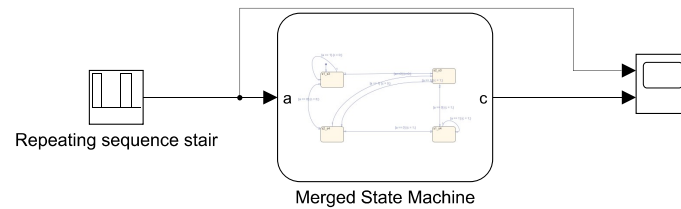


Figure 1: Simulink layout showing a Repeating sequence stair driving the Merged State Machine's input port 'a', with the signal 'a' and output port 'c' routed to a Scope for comparison[cite: 2].

Name: Srinjoy Chakraborty

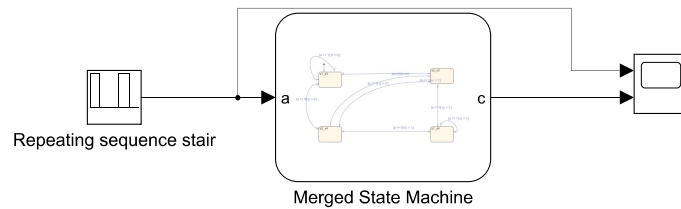
Indian Institute of Technology Kanpur
CS637 Embedded and Cyber-Physical SystemsRoll No: 251040638
e.g. 170001Dept.: EE
e.g. CSEHomework Assignment 1
Deadline: September 11, 2026

Figure 2: Mealy-style Stateflow chart implementing the mathematically derived synchronous transitions.

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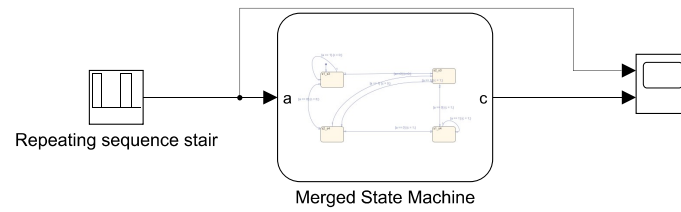
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e.g. 170001Dept.: EE
e.g. CSEHomework Assignment 1
Deadline: September 11, 2026

Figure 3: Scope plot demonstrating the dynamic output c correctly responding to input a based on the system's internal state history.

Name: Roll No:
e.g. 170001Dept.:
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CS637 Embedded and Cyber-Physical Systems

Homework Assignment 1

Deadline: September 11, 2026

Problem 6. (10 points)

Consider the following SDF model:



The numbers adjacent to the ports indicate the number of tokens produced or consumed by the actor when it fires. Answer the following questions about this model.

(a) Let q_A , q_B , and q_C denote the number of firings of actors A , B , and C , respectively. Write down the balance equations and find the least positive integer solution.

(b) Find a schedule for an unbounded execution that minimizes the buffer sizes on the two communication channels. What is the resulting size of the buffers?

Name: Srinjoy Chakraborty

Indian Institute of Technology Kanpur
CS637 Embedded and Cyber-Physical SystemsRoll No: 251040638
e.g. 170001Dept.: EE
e.g. CSE

Homework Assignment 1

Deadline: September 11, 2026

Problem 6: Synchronous Dataflow (SDF) Model**(a) Balance Equations and Firing Rates**

To find the firing rates, we establish balance equations for each communication channel to ensure the number of tokens produced equals the number of tokens consumed over one complete iteration.

- **Edge A to B:** Actor A produces 1 token per firing, and Actor B consumes 3 tokens.

$$1 \cdot q_A = 3 \cdot q_B$$

- **Edge B to C:** Actor B produces 2 tokens per firing, and Actor C consumes 3 tokens.

$$2 \cdot q_B = 3 \cdot q_C$$

Solving for q_A and q_C in terms of q_B :

$$q_A = 3q_B, \quad q_C = \frac{2}{3}q_B$$

To find the least positive integer solution, q_B must be a multiple of 3. Setting $q_B = 3$ yields:

$$q_A = 3(3) = 9, \quad q_C = \frac{2}{3}(3) = 2$$

The least positive integer solution is $q_A = 9$, $q_B = 3$, $q_C = 2$.

(b) Schedule and Buffer Sizes

To minimize buffer sizes, an actor is scheduled to fire as soon as it has the requisite tokens available. We track the tokens in Buffer AB (between A and B) and Buffer BC (between B and C):

1. Fire A 3 times: Buffer AB = 3, Buffer BC = 0.
2. Fire B 1 time: Buffer AB = 0, Buffer BC = 2.
3. Fire A 3 times, then fire B 1 time: Buffer AB = 0, Buffer BC = 4 (Previous 2 + New 2).
4. Fire C 1 time: Buffer AB = 0, Buffer BC = 1.
5. Fire A 3 times, then fire B 1 time: Buffer AB = 0, Buffer BC = 3 (Previous 1 + New 2).
6. Fire C 1 time: Buffer AB = 0, Buffer BC = 0.

Optimized Schedule: A, A, A, B, A, A, A, B, C, A, A, A, B, C

Resulting Buffer Sizes: Based on the peak token counts in the execution trace above, the minimum required size for the communication buffers are:

- Size of Buffer A $\rightarrow$ B: **3**
- Size of Buffer B $\rightarrow$ C: **4**